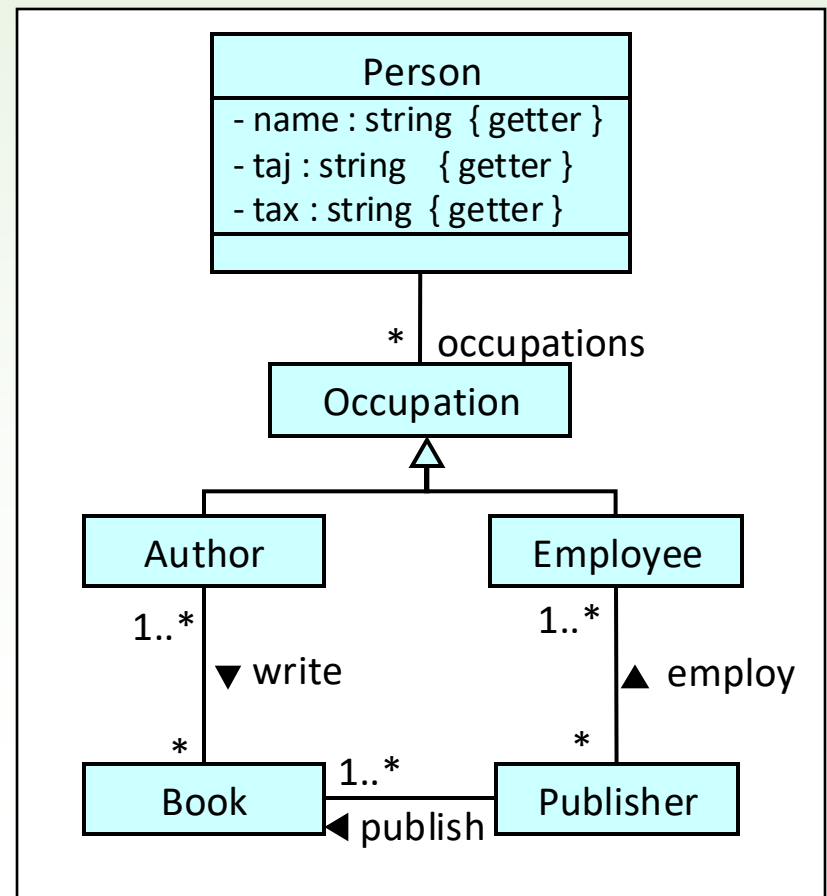
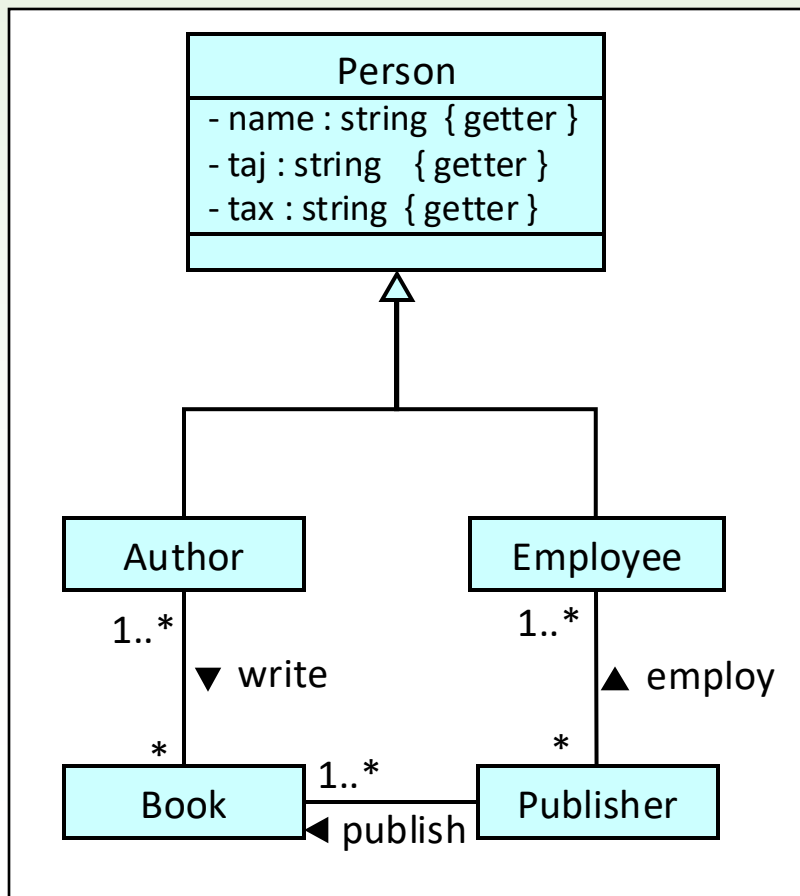


Book publishing

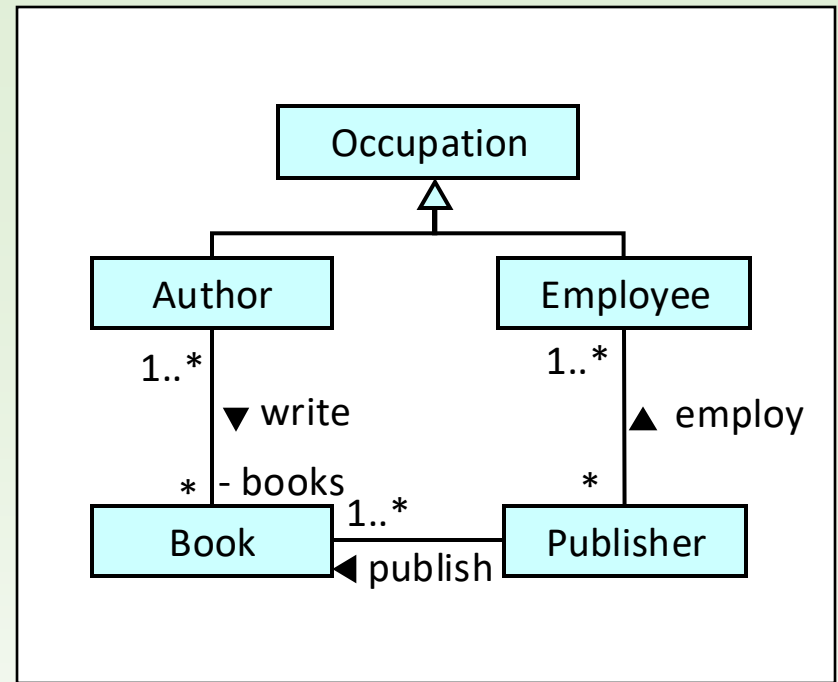
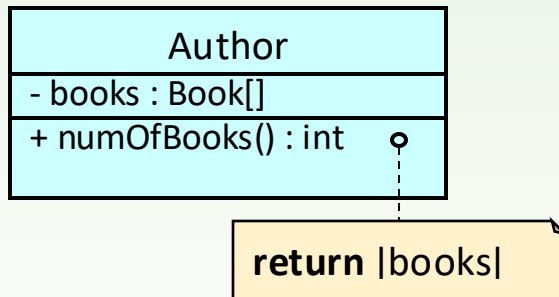
Books are written by at least one author and published by exactly one publisher. A publisher publishes at least one book. A publisher employs at least one employee. Both employees and authors have a name, social security number (TAJ card number), and tax identification number or code. It should be possible for someone to have multiple occupations at the same time.



Book publishing

Task: How many books has a given author written?

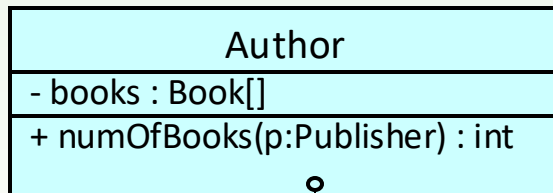
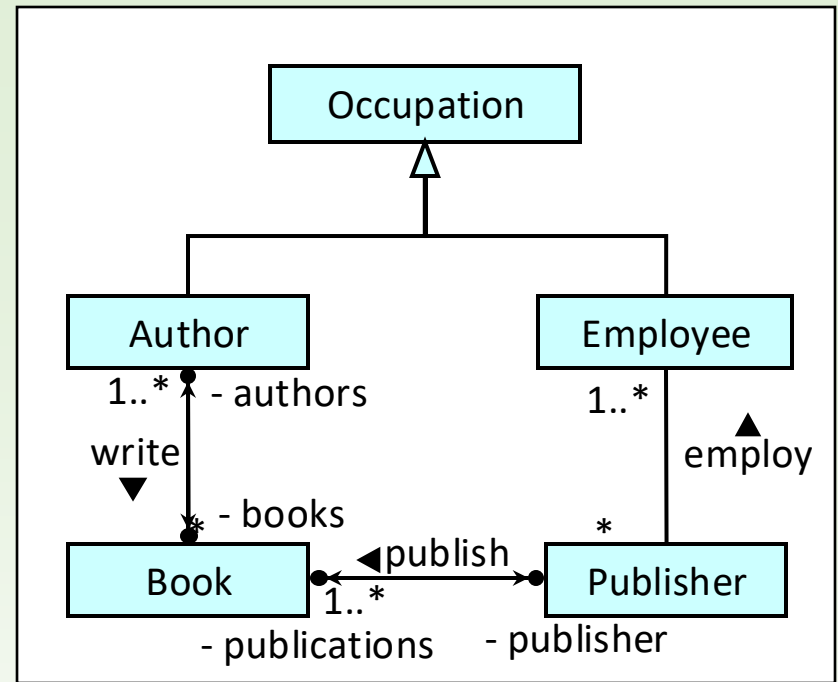
- Which class should numOfBooks() belong to?



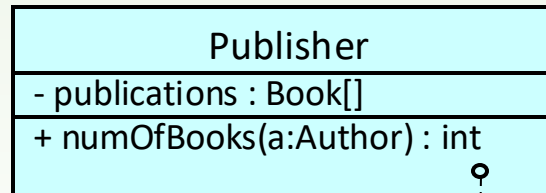
Book publishing

Task: How many books has a given author written for a given publisher?

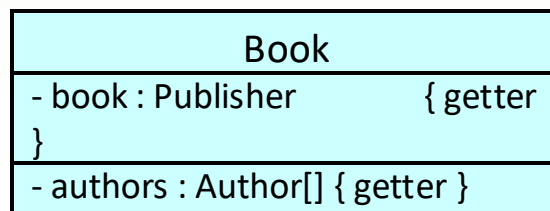
- Which class should numOfBooks() belong to?



return $\sum_{1} \text{book in books} \mid \text{book.publisher} = \text{kiad}$



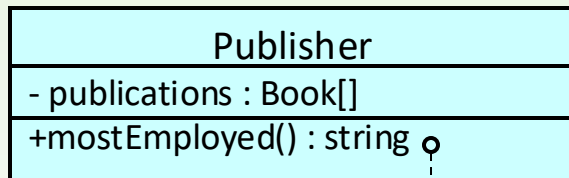
return $\sum_{1} \text{book in publications} \mid \sum_{\text{author in book.authors} \mid \text{author} = a}$



Book publishing

Task: Who is the most frequently employed author by a publisher?

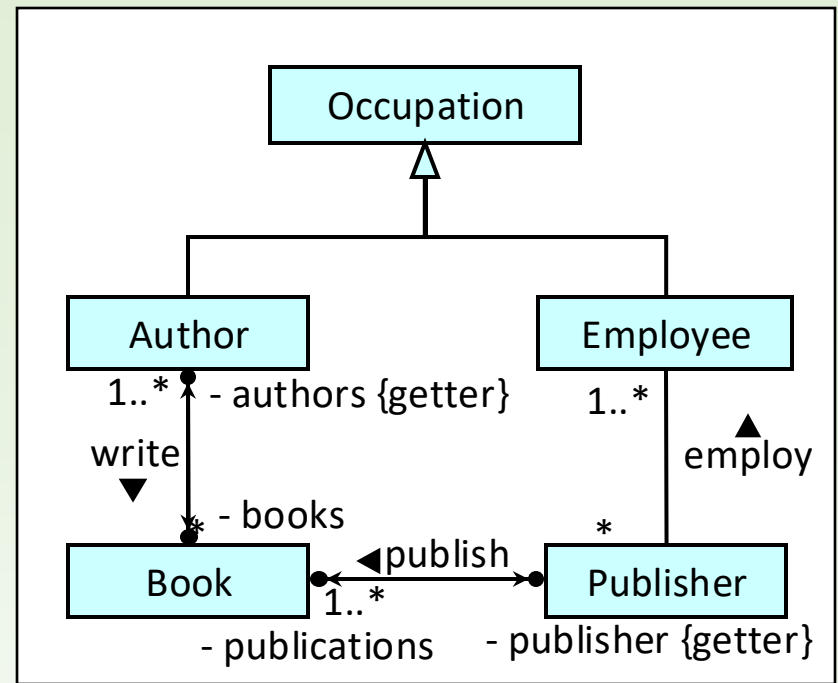
- Which class should `mostEmployed()` belong to?



(max, maxAuthor) = **MAX**_{book in publications and author in book.authors} author.numOfBooks(this))
return maxAuthor

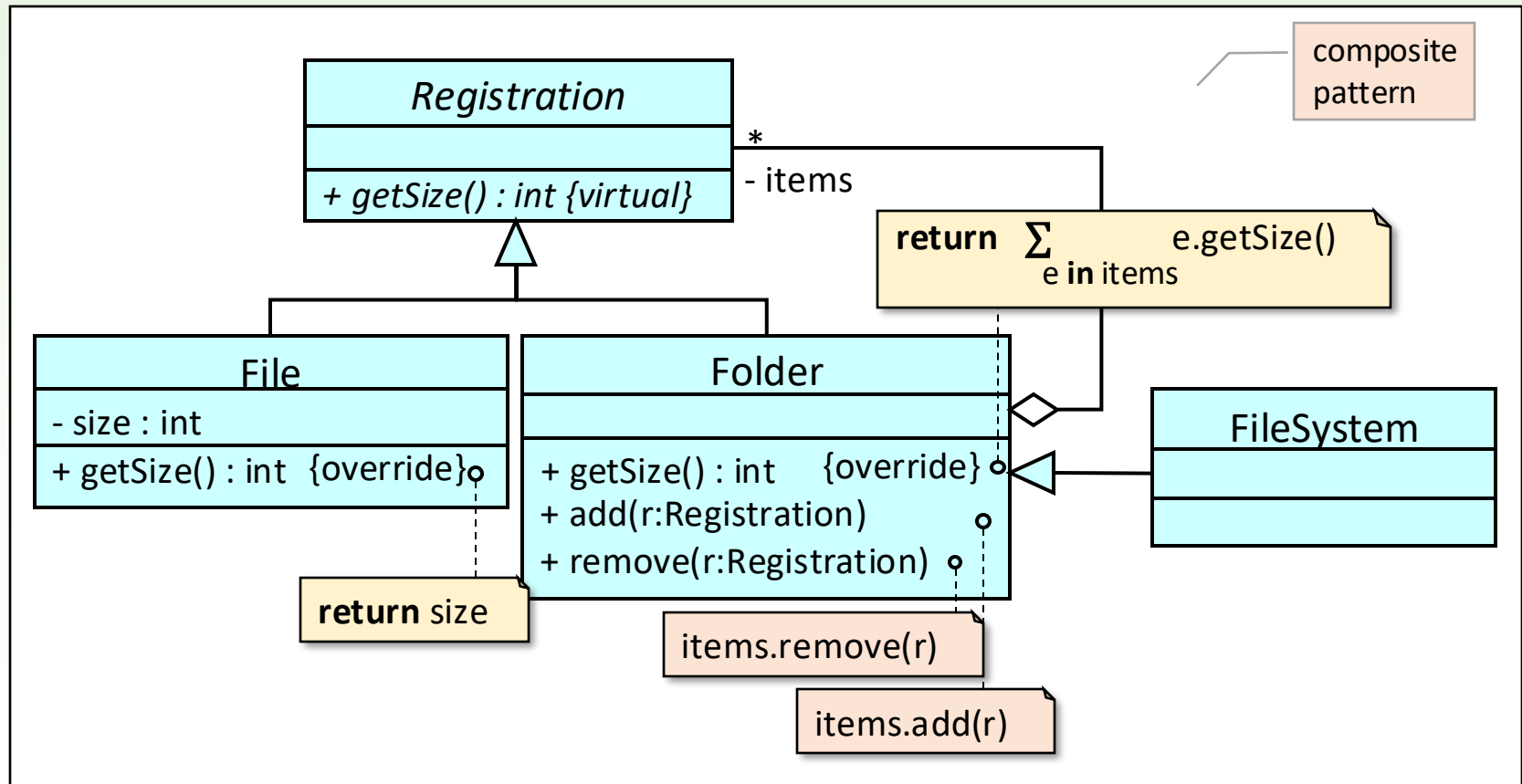
Could we come up with a more efficient solution?

Yes, by establishing a direct relation between Author and Publisher.



File system

In a computer file system, files are organized into folders. Files can belong directly to the file system or can be located within one of its folders. Each folder can contain any number of files or subfolders. For a file, the occupied storage space is known; the storage space of a folder is the sum of the storage spaces of the files and subfolders it contains. How much storage space does a given folder occupy?



Gardening

A garden is taken care of by a gardener. The garden consists of parcels, and each parcel can be planted with one type of plant. The time of planting is stored (in months). The plants can be vegetables, such as potatoes, peas, or onions; or flowers, such as tulips, carnations, or roses. The ripening time of the plants is also known (in months). The gardener should list the parcels where vegetables will ripen in a given month. (In other words: which parcels are harvestable?)

